235 post midterm tutorial

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1

1.1

Let d divide a, b. Then d divides ra, sb, hence their sum.

1.2

WLOG let $a, b > 0$. Let $t = ra + sb$ be the smallest strictly positive integer linear combination of a, b. We note that $t \leq a$ by setting $r = 1, s = 0$ so write $a = qt + p$, $0 \leq p < t$.

$$\begin{aligned} p &= a - pt \\ &= a - pra - psb \\ &= (1 - pr)a - psb \end{aligned}$$

If this is non-zero, it contradicts the minimality of t, so it must be 0. So $t|a$ and similarly $t|b$.

1.3

Since every common divisor of a, b divide t (part 1) and t is a common divisor of a, b we see it is the greatest such common divisor.

2

We induct on c. Base case: $c = 0$:

$$(a + b) + 0 = a + b = a + (b + 0)$$

We use the inductive definition above to say $x = x + 0$ in the above. Assume the identity holds for $c = n$. Then:

$$(a + b) + S(n) = S(a + b + n) \text{ (inductive definition)}$$

By the induction hypothesis this is equal to:

$$\begin{aligned} S(a + (b + n)) &= a + S(b + n) \text{ (inductive definition)} \\ &= a + (b + S(n)) \text{ (inductive definition)} \end{aligned}$$

and we see the inductive case holds.

3

3.1

$a^{p-1} \equiv 1 \bmod p$.

3.2

Note that Fermat's says $3^{100} \equiv 1 \bmod 101$ (as 101 prime and $101 \nmid 3$). Write $n = n_0 + 10n_1 + 100k$ where n_0, n_1 are the 1s and 10s place of n (the last two digits) and the rest is divisible by 100 (in other words n mod $100 \equiv n_0 + 10n_1$ which are the last 2 digits of n). Then:

$$\begin{aligned} 3^n &= 3^{n_0+10n_1+100k} \\ &= 3^{n_0+10n_1} \cdot 3^{100k} \\ &= 3^{n_0+10n_1} \cdot (3^{100})^k \\ &\equiv (3^{n_0+10n_1} \cdot 1^k) \bmod 100 \\ &= 3^{n_0+10n_1} \end{aligned}$$

So we see 3^n depends only on the last two digits as needed. Then the example given is congruent to 3^2 mod 101. We see the answer is 9 mod 101.

4

4.1

Let $x^2 = 1$. Then $x^2 - 1 = 0$ or $(x-1)(x+1) = 0$. Since $\mathbb{Z}/p\mathbb{Z}$ is a field, it has no zero divisors. So solutions to this must have $x - 1$ or $x + 1$ be zero. In particular, $x = 1$ or $x = -1$. Since p is odd, $p > 2$ so $1 \neq -1$ and we have exactly 2 solutions.

4.2

We note that $(a^{\frac{p-1}{2}})^2 = a^{p-1}$. By Fermat's, this is 1 (since p prime and $a \not\equiv 0 \bmod p$). Then the part above says that $a^{\frac{p-1}{2}} = \pm 1$